Universal Simulator

Documentation

Version 3.0.34

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2 Module: boxes

2.1 Class: Accumulator

2.1.1 Inputs

Parameter	Type	Default	Description
`initial`	`double`	`0.0`	Initial `value` at reset
`change`	`double`	`0.0`	Change added to `value`
`minValue`	`double`	$-\infty$	Minimum possible `value`
`maxValue`	`double`	∞	Maximum possible `value`

2.1.2 Outputs

Parameter	Type	Default	Description
`value`	`double`	`0.0`	Accumulated value
`signal`	`double`	`0.0`	Synonym for `value`
`isAtMin`	`bool`	`FALSE`	Is `value` equal to `minValue`?
`isAtMax`	`bool`	`FALSE`	Is `value` equal to `maxValue`?

2.1.3 Usage

The `Accumulator` box keeps track of an accumulated sum. For every simulation step, change (can be negative) is added to its `value`. This boxscript calculates the day-degrees above a threshold of 10 °C (a DayDegrees box would allow you more control with less BoxScript code):

```
// accumulator1.box
Simulation sim {
  Calendar calendar {
    .begin = 01/05/2009
    .end   = 30/09/2009
  }
  Records weather {
    .fileName = "weather/flakkebjerg 2009.txt"
  }
  Accumulator dd18 {
    .change = if (weather[Tavg] > 18) then weather[Tavg] - 18 else 0
  }
  OutputR {
    PageR {
      .xAxis = calendar[date]
      PlotR {
        .layout = "merged"
        .ports = weather[Tavg] | dd18[change] | dd18[value]
      }
    }
  }
}
```

Note that in the output, if a port is named "value" then it is by default labelled after its parent name (here "dd18"):

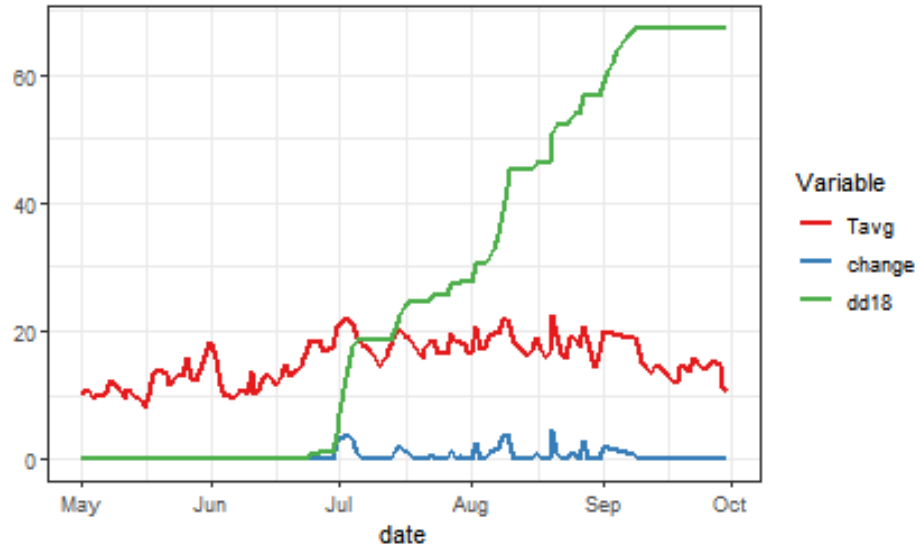


Figure 1: Accumulator example

2.2 Class: AhFromRh

2.2.1 Inputs

Parameter	Type	Default	Description
`temperature`	`double`	`0.0`	Temperature
`rh`	`double`	`0.0`	Relative humidity
`value`	`double`	`0.0`	Absolute humidity

2.3 Class: AllTypes

2.3.1 Inputs

Parameter	Type	Default	Description
`bool`	`bool`	`FALSE`	
`int`	`int`	`0`	
`double`	`double`	`0.0`	
`string`	`string`		
`date`	`date`	`null`	
`time`	`time`	`null`	
`datetime`	`datetime`	`null`	
`baredate`	`baredate`	`null`	
`path`	`path`	`null`	
`vec_bool`	`vec_bool`	`null`	
`vec_int`	`vec_int`	`null`	
`vec_double`	`vec_double`	`null`	
`vec_string`	`vec_string`	`null`	
`vec_date`	`vec_date`	`null`	
`vec_time`	`vec_time`	`null`	
`vec_datetime`	`vec_datetime`	`null`	
`vec_baredate`	`vec_baredate`	`null`	

2.4 Class: Briere

2.4.1 Inputs

Parameter	Type	Default	Description
`T`	`double`	a)	
`timeStepDays`	`double`	b)	
`resetTotal`	`bool`	`FALSE`	Reset `total` zero?
`isTicking`	`bool`	`TRUE`	Is the physiological time running? Else `step` will be zero
`T0`	`double`	`0.0`	Lower temperature threshold
`Tmax`	`double`	`0.0`	Upper temperature threshold
`a`	`double`	`0.0`	Scaling parameter

a) `weather[Tavg]`
b) `calendar[timeStepDays]`

2.4.2 Outputs

Parameter	Type	Default	Description
`step`	`double`	`0.0`	Increment in physiological time units (PTU)
`total`	`double`	`0.0`	Total time since reset

2.4.3 Background

The `Briere` box implements the non-linear model for physiological development of Briere et al. (1999) defined by three inputs: `T0`, `Tmax` and `a`. They designate the lower and upper threshold for development, respectively, and a scaling parameter with no direct biological meaning. The equation for development rate is

$$\frac{1}{L} = aT(T - T_0)\sqrt{T_{\max} - T}$$

The position of the temperature optimum is given implicitly by T_0 and $T_{\max}$ and can be derived from the above equation as

$$T_{\text{opt}} = \frac{\sqrt{9T_0^2 - 16T_0T_{\max} + 16T_{\max}^2} + 3T_0 + 4T_{\max}}{10}$$

Maybe of more relevance is the finding, that if you want to define the curve by its optimum rather than its maximum temperature, you can calculate the maximum temperature from the optimal temperature by

$$T_{\max} = \frac{T_{\text{opt}}(5T_{\text{opt}} - 3T_0)}{4T_{\text{opt}} - 2T_0}$$

2.4.4 Parameter estimation

The three parameters of the curve can be estimated by non-linear regression. Here is a typical data set for temperature-dependent insect development rate. It shows the development time (L) and rate ($1/L$) at different temperatures (T) for brown marmorated stink bug (*Halyomorpha halys*) (from Nielsen et al. (2008)):

T [°C]	L [d]	$\frac{1}{L}$ [d^{-1}]
15	∞	0.0000
17	121.50	0.0082
20	81.07	0.0123
25	44.65	0.0224

27	36.67	0.0273
30	33.73	0.0296
33	37.67	0.0265
35	∞	0.0000

Using R *nls* we arrive at the estimates, $T_0 = 14.2$ °C, $T_{\max} = 35.0$ °C and $a = 2.836 \times 10^{-5}$ d^{-1} that we can plug into this boxscript, which also shows two alternative models (simple linear and the DayDegrees box, which features a breakpoint at the optimum temperature):

```
// briere1.box
Simulation sim {
  .steps = temperature[steps]
  SequenceByStep temperature {
    .min = 12
    .max = 37
    .by = 0.2
  }
  DayDegrees dayDegrees {
    .timeStepDays = 1
    .T0 = 14.2
    .Topt = 29
    .Tmax = 35
    .T = temperature[value]
  }
  Briere briere {
    .timeStepDays = 1
    .T0 = 14.2
    .Tmax = 35
    .a = 2.836e-5
    .T = temperature[value]
  }
  OutputR {
    .scripts = "briere1.R"
    OutputText {
      .ports = temperature[value] | dayDegrees[step] | briere[step]
    }
  }
}
```

This is the plot produced:

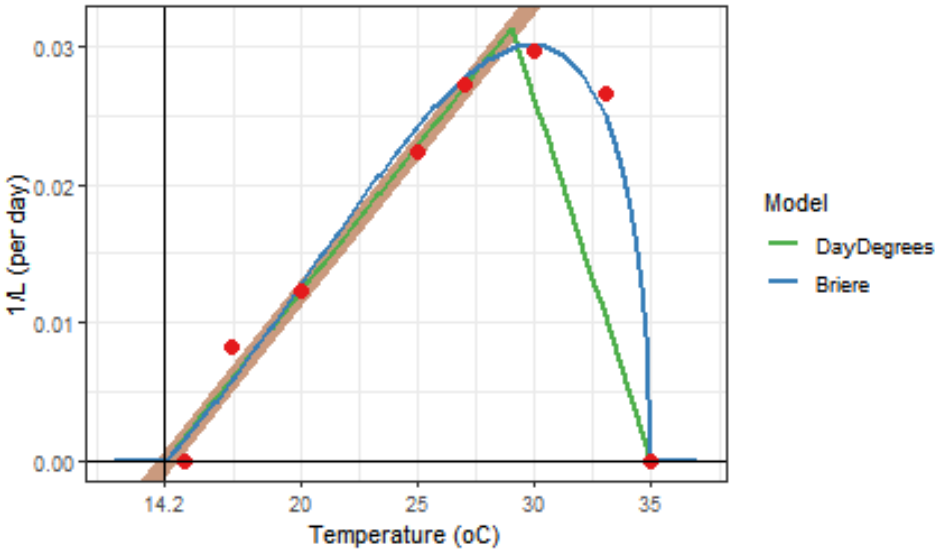


Figure 2: Briere development-rate example

2.4.5 Usage

In practical usage, you often want to turn the ticking of the physiological clock on and off. Sometimes, you may want to reset it to zero too. This is what the two remaining input ports (`isTicking` and `resetTotal`) are for. They are both of the boolean type. In this script, the clock is only ticking in the period 1 May to 1 October. Moreover, it is reset to zero on 1 May and 1 July:

```
// briere2.box
Simulation sim {
  Calendar calendar {
    .begin = 01/01/2009
    .end   = 31/12/2010
  }
  Records weather {
    .fileName = "weather/flakkebjerg 2009.txt"
    .cycle = TRUE
  }
  Briere clock {
    .isTicking = (calendar[date] >= 01/05) && (calendar[date] < 01/10)
    .resetTotal = (calendar[date] == 01/05) || (calendar[date] == 01/07)
    .T0 = 8
    .Tmax = 32
    .a = 2.836e-5
  }
  OutputR {
    PageR {
      .xAxis = calendar[date]
      PlotR {
        .ports = clock[output::*] | clock[isTicking] | clock[resetTotal]
        .ggplot = "scale_x_datetime(breaks=date_breaks('3 months'),
                                    labels = date_format ('%b')) +
                   labs(x='2009 - 2010')"
      }
    }
  }
}
```

Noticable BoxScript tricks include

- the use of the `cycle` port in `Records` to recycle the weather data;
- the use of *bare dates* to ignore the year of the date (e.g., `01/05` to represent 1 May any year);
- the use of the `ggplot` port to `PlotR` to add details to the plot produced.

The output shows two years worth of data, identical since the weather of 2009 is used for both years:

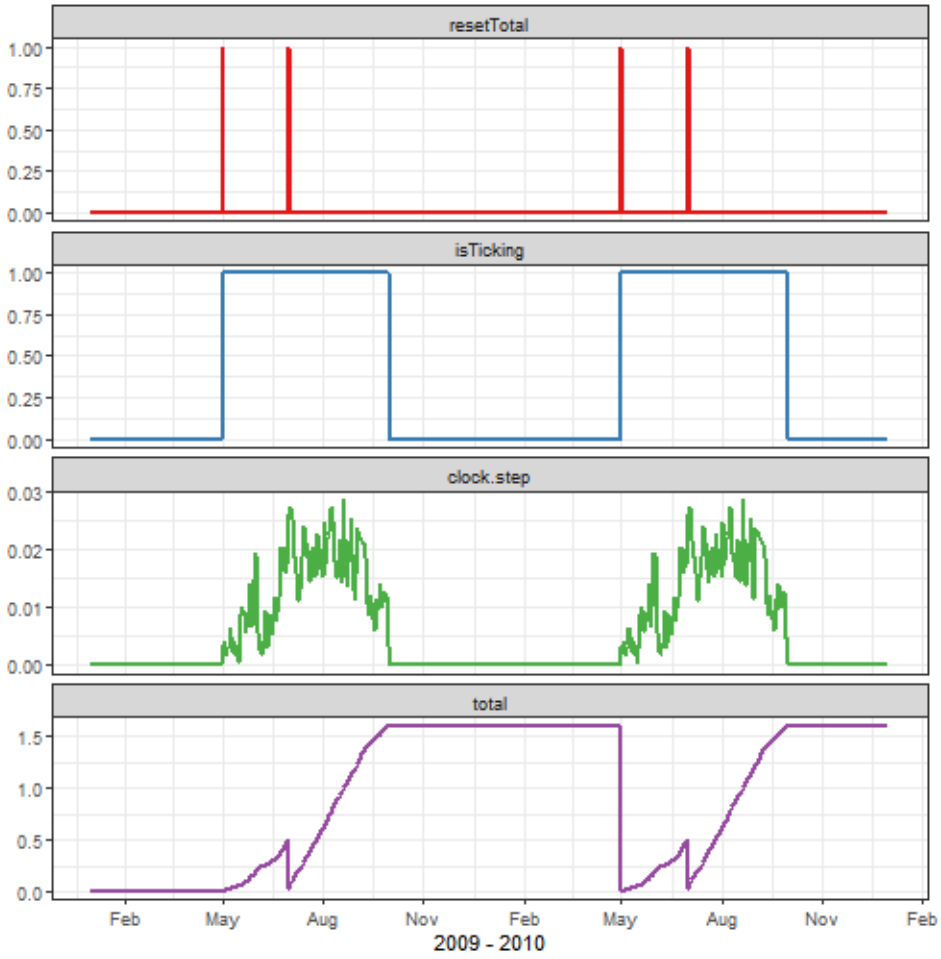


Figure 3: Briere clock switched on, off and reset

2.5 Class: Buffer

2.5.1 Inputs

Parameter	Type	Default	Description
`initial`	`double`	`0.0`	Initial amount at first place in buffer
`input`	`double`	`0.0`	Input to next place in buffer
`increment`	`double`	`0.0`	Increment added to every place in buffer
`size`	`int`	`100`	Number of places in buffer

2.5.2 Outputs

Parameter	Type	Default	Description
`sum`	`double`	`0.0`	Total amount in buffer
`average`	`double`	`0.0`	Average amount of places used in buffer
`buffer`	`vec_double`	`null`	The buffer itself

2.6 Class: Calendar

2.6.1 Inputs

Parameter	Type	Default	Description
latitude	double	52.0	Latitude
longitude	double	11.0	Longitiude
timeZone	int	1	Time zone
begin	datetime	a)	When simulation begins
end	datetime	null	When simulation ends (optional)
timeStep	int	1	Time step in units of timeUnit
timeUnit	string	"d"	Unit of time step (y|d|h|m|s)

a) 2000/01/01T00:00:00

2.6.2 Outputs

Parameter	Type	Default	Description
steps	int	0	Number of steps from begin to end
stepWithinDay	int	0	Step number within day
stepsPerDay	int	0	Number of steps per day
date	date	null	Current date
time	time	null	Current time of the day
dateTime	datetime	null	Current date and time
atMidnight	bool	FALSE	Just past midnight?
timeStepSecs	double	0.0	Time step duration
timeStepDays	double	0.0	Time step duration
totalTimeSteps	int	0	Total number of time steps since calendar was reset
totalTime	int	0	Total time since calendar was reset; in units of timeUnit
totalDays	double	0.0	Total time since calendar was reset in days
dayOfYear	int	0	Julian day
bareDate	baredate	null	date as a bare date

2.7 Class: Combinations

2.7.1 Outputs

Parameter	Type	Default	Description
iterations	int	0	Total number of combinations

2.8 Class: Computation

2.8.1 Outputs

Parameter	Type	Default	Description
step	string	construct	Name of current computation step

2.9 Class: Date

2.9.1 Inputs

Parameter	Type	Default	Description
`dmy`	`string`		A date string
`day`	`int`	`1`	Day of month
`month`	`int`	`1`	Month
`year`	`int`	`2000`	Year

2.9.2 Outputs

Parameter	Type	Default	Description
`date`	`date`	`null`	The date constructed
`value`	`date`	`null`	Synonym for `date`

2.10 Class: DateSplit

2.10.1 Inputs

Parameter	Type	Default	Description
`date`	`date`	`null`	The date to split

2.10.2 Outputs

Parameter	Type	Default	Description
`day`	`int`	`0`	Day of month
`month`	`int`	`0`	Month
`year`	`int`	`0`	Year

2.11 Class: DateTime

2.11.1 Inputs

Parameter	Type	Default	Description
`dateTimeString`	`string`		A datetime string
`day`	`int`	`1`	Day of month
`month`	`int`	`1`	Month
`year`	`int`	`2000`	Year
`hour`	`int`	`0`	Hour of the day
`minute`	`int`	`0`	Minute of the hour
`second`	`int`	`0`	Second of the minute

2.11.2 Outputs

Parameter	Type	Default	Description
`dateTime`	`datetime`	`null`	The datetime constructed
`value`	`datetime`	`null`	Synonym for `dateTime`

2.12 Class: DateTimeSignal

2.12.1 Inputs

Parameter	Type	Default	Description
initialSignal	double	0.0	Value of signal at reset
initialFlag	bool	FALSE	Value of flag at reset
currentDateTime	datetime	a)	
beginDate	baredate	1/1	The beginning of the date interval
endDate	baredate	31/12	The end of the date interval
beginTime	time	00:00:00	The beginning of the time interval
endTime	time	00:00:00	The end of the time interval
signalInside	double	1.0	Resulting signal when day and time are inside the day and time intervals
signalOutside	double	0.0	Resulting signal when day and time are outside the day and time intervals
circadian	bool	TRUE	Tells whether signal follows a daily rythm

a) calendar[dateTime]

2.12.2 Outputs

Parameter	Type	Default	Description
signal	double	0.0	Value of the signal
flagIsUp	bool	FALSE	Is the signal on?
flagJustRaised	bool	FALSE	Was the signal just set on?
flagJustLowered	bool	FALSE	Was the signal just set off?
value	double	0.0	Synonym for signal

2.12.3 Usage

The DateTimeSignal box sets the signal to signalInside when the currentDateTime is inside the set limits, otherwise it is set to signalOutside. The limits are set by date (beginDate and endDate) and time of the day (beginTime and endTime) separately. Note that beginDate and endDate are *bare dates*, which means that they just specify the day and month, not the year. This means that you can have a calendar interval across 1 January, e.g. from 1 December (beginDate) to 1 March (endDate). All intervals of dates and times are closed (i.e. limits are inclusive). If beginTime and endTime keep their default value (00:00), the time of day is ignored.

If circadian is TRUE then currentDateTime is considered inside the limits if the date of currentDateTime is inside the [beginDate; endDate] period, and the time of currentDateTime is inside the [beginTime; endTime] period (specifying a recurring signal).

If circadian is FALSE then currentDateTime is considered inside the limits if it falls inside the period from beginTime on beginDate to endTime on endDate (specifying a single signal).

2.12.4 Circadian signals

The difference that circadian makes is easier to understand by example. First we demonstrate a circadian signal:

```
// date_time_signal1.box
Simulation sim {
  Calendar calendar {
    .begin = 01/01/2001T12:00
    .end   = 10/01/2001T12:00
```

```
    .timeStep = 15
    .timeUnit = "m"
  }
  DateTimeSignal signal {
    .beginDate  = 03/01/2001
    .endDate    = 07/01/2001
    .beginTime  = 14:00
    .endTime    = 20:00
    .initialSignal = 50
    .signalInside  = 80
    .signalOutside = 50
  }
  OutputR {
    PageR {
      .xAxis = calendar[dateTime]
      PlotR {
        .ports = signal[value] | signal[flagIsUp] |
                 signal[flagJustRaised] | signal[flagJustLowered]
        .ggplot = "scale_x_datetime(breaks=date_breaks('days'),
                                    labels = date_format ('%d')) +
                   labs(x='January')"
      }
    }
  }
}
```

The boxscript raises the signal for five days between 14:00 and 20:00 every day, as shown in the output:

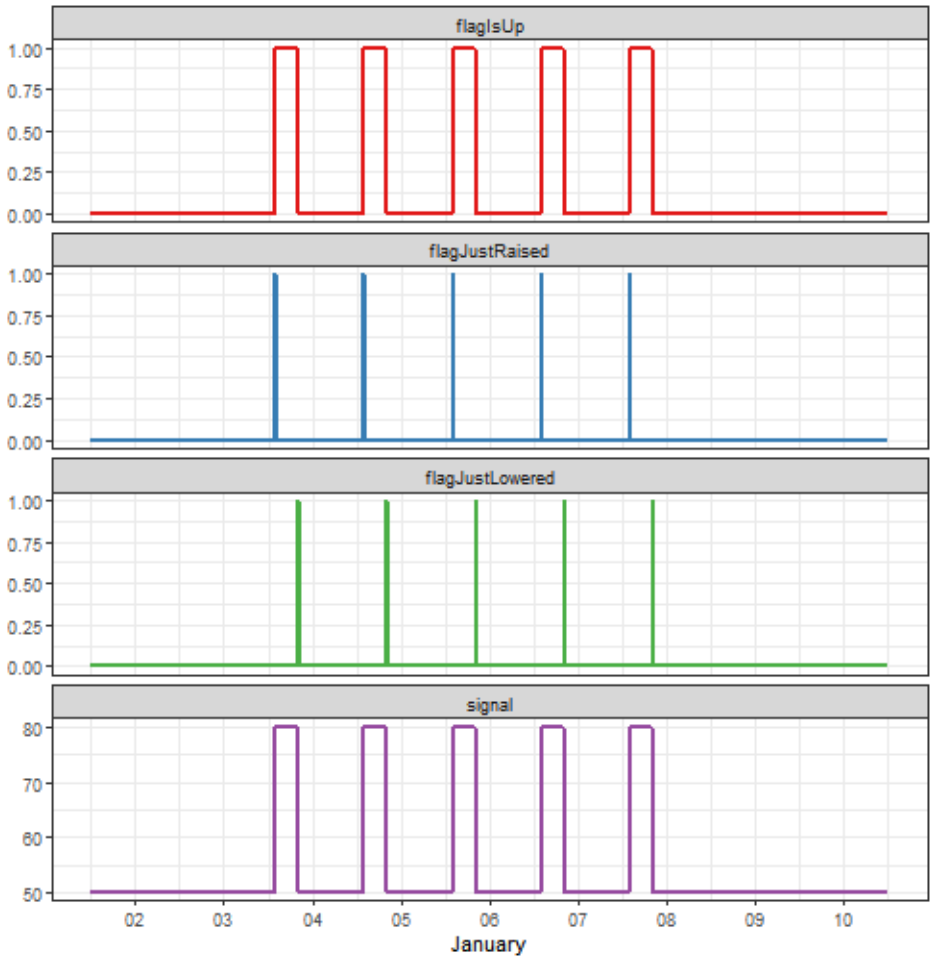


Figure 4: Circadian DateTimeSignal

You can check the precision of the signal going *on* at the Universal Simulator prompt with the `rows 295 300` command:

```
> rows 295 300
...
           dateTime flagIsUp flagJustRaised flagJustLowered signal iteration step
2001/01/04T13:30:00        0              0               0   50.0         1  294
2001/01/04T13:45:00        0              0               0   50.0         1  295
2001/01/04T14:00:00        1              1               0   80.0         1  296
2001/01/04T14:15:00        1              0               0   80.0         1  297
2001/01/04T14:30:00        1              0               0   80.0         1  298
2001/01/04T14:45:00        1              0               0   80.0         1  299
```

Or at the R prompt with `sim[295:300,]`:

```
> sim[295:300,]
              dateTime flagIsUp flagJustRaised flagJustLowered signal iteration step
295 2001-01-04 13:30:00        0              0               0     50         1  294
296 2001-01-04 13:45:00        0              0               0     50         1  295
297 2001-01-04 14:00:00        1              1               0     80         1  296
298 2001-01-04 14:15:00        1              0               0     80         1  297
299 2001-01-04 14:30:00        1              0               0     80         1  298
300 2001-01-04 14:45:00        1              0               0     80         1  299
```

Similarly, you can check the timeliness of the signal going *off*:

```
> rows 320 325
...
           dateTime flagIsUp flagJustRaised flagJustLowered signal iteration step
```

```
2001/01/04T19:45:00        1              0              0    80.0         1  319
2001/01/04T20:00:00        1              0              0    80.0         1  320
2001/01/04T20:15:00        0              0              1    50.0         1  321
2001/01/04T20:30:00        0              0              0    50.0         1  322
2001/01/04T20:45:00        0              0              0    50.0         1  323
2001/01/04T21:00:00        0              0              0    50.0         1  324
```

2.12.5 Non-circadian signals

In the next script (date_time_signal2.box), we only changed one detail, namely the value of `circadian` in the `signal` box:

```
DateTimeSignal signal {
  .beginDate  = 03/01/2001
  .beginTime  = 14:00
  .endDate    = 07/01/2001
  .endTime    = 20:00
  .initialSignal = 50
  .signalInside  = 80
  .signalOutside = 50
  .circadian = FALSE
}
```

We also re-arranged the ordering of the date and time inputs to make their meaning more clear. The output shows that the signal is raised only through one time period stretching 4 days and 6 hours:

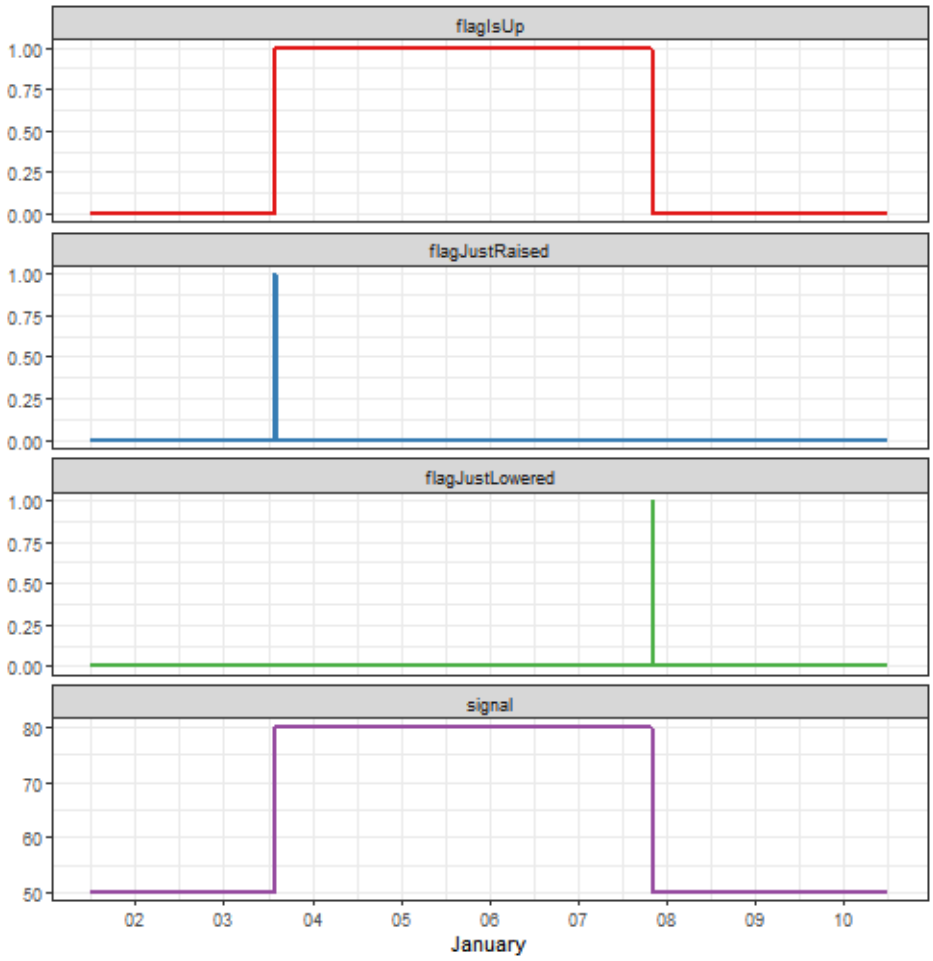


Figure 5: Non-circadian DateTimeSignal

The precise timing is readily revealed:

```
> rows 199 202
...
```

```
           dateTime flagIsUp flagJustRaised flagJustLowered signal iteration step
2001/01/03T13:30:00        0              0               0   50.0         1  198
2001/01/03T13:45:00        0              0               0   50.0         1  199
2001/01/03T14:00:00        1              1               0   80.0         1  200
2001/01/03T14:15:00        1              0               0   80.0         1  201

> rows 608 611
...
           dateTime flagIsUp flagJustRaised flagJustLowered signal iteration step
2001/01/07T19:45:00        1              0               0   80.0         1  607
2001/01/07T20:00:00        1              0               0   80.0         1  608
2001/01/07T20:15:00        0              0               1   50.0         1  609
2001/01/07T20:30:00        0              0               0   50.0         1  610
```

Alternatively, we can ignore the time of day all together, as shown in the date_time_signal3.box script:

```
DateTimeSignal signal {
  .beginDate  = 03/01/2001
  .endDate    = 07/01/2001
  .initialSignal = 50
  .signalInside  = 80
  .signalOutside = 50
  .circadian = FALSE
}
```

This gives us five full days. Here the signal is going *on* at midnight:

```
> rows 143 146
...
           dateTime flagIsUp flagJustRaised flagJustLowered signal iteration step
2001/01/02T23:30:00        0              0               0   50.0         1  142
2001/01/02T23:45:00        0              0               0   50.0         1  143
2001/01/03T00:00:00        1              1               0   80.0         1  144
2001/01/03T00:15:00        1              0               0   80.0         1  145
```

And here the signal is going *off* at midnight:

```
> rows 623 626
...
           dateTime flagIsUp flagJustRaised flagJustLowered signal iteration step
2001/01/07T23:30:00        1              0               0   80.0         1  622
2001/01/07T23:45:00        1              0               0   80.0         1  623
2001/01/08T00:00:00        0              0               1   50.0         1  624
2001/01/08T00:15:00        0              0               0   50.0         1  625
```

2.13 Class: DayDegrees

2.13.1 Inputs

Parameter	Type	Default	Description
`T`	`double`	a)	
`timeStepDays`	`double`	b)	
`resetTotal`	`bool`	`FALSE`	Reset `total` zero?
`isTicking`	`bool`	`TRUE`	Is the physiological time running? Else `step` will be zero
`T0`	`double`	`0.0`	Lower temperature threshold
`Topt`	`double`	`100.0`	Optimum temperate; linear decline from here to `Tmax`
`Tmax`	`double`	`100.0`	Upper temperature threshold

a) `weather[Tavg]`
b) `calendar[timeStepDays]`

2.13.2 Outputs

Parameter	Type	Default	Description
`step`	`double`	`0.0`	Increment in day-degrees
`total`	`double`	`0.0`	Total time since reset

2.14 Class: DemandBudget

2.14.1 Inputs

Parameter	Type	Default	Description
`demandNet`	`double`	`0.0`	Net demand
`demandResp`	`double`	`0.0`	Respiration demand
`egested`	`double`	`0.0`	Egested or unused fraction of prey
`conversionCost`	`double`	`0.0`	Cost of converting prey into consumer)

2.14.2 Outputs

Parameter	Type	Default	Description
`demandGross`	`double`	`0.0`	Gross demand including all expenditures

2.15 Class: DewPoint

2.15.1 Inputs

Parameter	Type	Default	Description
`temperature`	`double`	`0.0`	Air temperature
`rh`	`double`	`0.0`	Air relative humidity

2.15.2 Outputs

Parameter	Type	Default	Description
`value`	`double`	`0.0`	Dew point temperature

2.16 Class: Distribution

2.16.1 Inputs

Parameter	Type	Default	Description
`input`	`double`	`0.0`	the input port to anayse
`divisor`	`double`	`100.0`	Divisor applied to `sections
`sections`	`vec_int`	a)	Sections applied at the end of every iteration
`steps`	`int`	`/.[steps]`	

a) `c(1,5,25,50,75,95,99)`

2.17 Class: FixedSignal

2.17.1 Inputs

Parameter	Type	Default	Description
initialSignal	double	0.0	Value of signal at reset
initialFlag	bool	FALSE	Value of flag at reset

2.17.2 Outputs

Parameter	Type	Default	Description
signal	double	0.0	Value of the signal
flagIsUp	bool	FALSE	Is the signal on?
flagJustRaised	bool	FALSE	Was the signal just set on?
flagJustLowered	bool	FALSE	Was the signal just set off?
value	double	0.0	Synonym for signal

2.18 Class: FoodWeb

2.18.1 Inputs

Parameter	Type	Default	Description
timeStep	double	1.0	Simulation time step

2.19 Class: FoodWebCalcParasiteHost

2.19.1 Inputs

Parameter	Type	Default	Description
timeStep	double	.. [timeStep]	

2.20 Class: FoodWebCalcPredatorPrey

2.20.1 Inputs

Parameter	Type	Default	Description
timeStep	double	.. [timeStep]	

2.21 Class: FunctionalResponse

2.21.1 Inputs

Parameter	Type	Default	Description
attacker	double	0.0	Attacker density
prey	double	0.0	Resource density
demandGross	double	0.0	Gross demand of attacker
attackRate	double	0.0	#attacks per #prey per #predator per time unit
timeStep	double	1.0	Time step

2.21.2 Outputs

Parameter	Type	Default	Description
supplyGross	double	0.0	Gross supply (number of prey attacked)
propPreyAttacked	double	0.0	The proportion of prey attacked
searchEfficacy	double	0.0	Efficacy in finding prey [0;1]
sdRatioGross	double	0.0	Predator's gross supply/demand ratio

2.22 Class: Host

2.22.1 Inputs

Parameter	Type	Default	Description
density	double	0.0	Density
attackRate	double	0.0	Attack rate
gainFactor	double	1.0	Conversion of resource into supply units

2.22.2 Outputs

Parameter	Type	Default	Description
propAttacked	double	0.0	Proportion of hosts attacked
numAttacked	double	0.0	Number of hosts attacked

2.23 Class: Hump

2.23.1 Inputs

Parameter	Type	Default	Description
x	double	0.0	x-value at which to calculate curve value
x0	double	0.0	Beginning of curve
x1	double	1.0	End of curve
ymin	double	0.0	Minimum value of curve
ymax	double	1.0	Maximum value of curve
ditch	bool	FALSE	Is the curve a ditch?

2.23.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Curve value at x-value

2.24 Class: IgnoredBox

2.25 Class: LobryRossoFlandrois

2.25.1 Inputs

Parameter	Type	Default	Description
T	double	a)	
timeStepDays	double	b)	
resetTotal	bool	FALSE	Reset total zero?

Parameter	Type	Default	Description
isTicking	bool	TRUE	Is the physiological time running? Else step will be zero
Tmin	double	0.0	Lower temperature threshold
Topt	double	0.0	Optimal temperature
Tmax	double	0.0	Upper temperature threshold
maximum	double	0.0	Maximum development rate

a) weather[Tavg]
b) calendar[timeStepDays]

2.25.2 Outputs

Parameter	Type	Default	Description
step	double	0.0	Increment in physiological time units (PTU)
total	double	0.0	Total time since reset

2.26 Class: Maker

2.26.1 Inputs

Parameter	Type	Default	Description
replicates	int	1	Supply either, (1) the number of clones to replicate from child boxes;
names	vec_string	null	or (2) the names of clones;
fileName	string		or (3) a file; each line specifies a clone; column headings label inputs to each clone.

2.27 Class: Maximum

2.27.1 Inputs

Parameter	Type	Default	Description
values	vec_double	null	Input values

2.27.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Maximum of inputs; zero if no inputs
count	int	0	Number of inputs

2.28 Class: MaximumAt

2.28.1 Inputs

Parameter	Type	Default	Description
step	int	/.[step]	
track	double	0.0	Value to track for max value

2.28.2 Outputs

Parameter	Type	Default	Description
`at`	`int`	`0`	Step in which maximum value was attained
`value`	`double`	`0.0`	The maximum value attained

2.29 Class: Mean

2.29.1 Inputs

Parameter	Type	Default	Description
`values`	`vec_double`	`null`	Input values

2.29.2 Outputs

Parameter	Type	Default	Description
`value`	`double`	`0.0`	Average of inputs; zero if no inputs
`count`	`int`	`0`	Number of inputs

2.30 Class: Message

2.30.1 Inputs

Parameter	Type	Default	Description
`text`	`string`		Text message to print
`when`	`string`		In which step to print the message (reset, initialize, etc.)

2.31 Class: Minimum

2.31.1 Inputs

Parameter	Type	Default	Description
`values`	`vec_double`	`null`	Input values

2.31.2 Outputs

Parameter	Type	Default	Description
`value`	`double`	`0.0`	Minimum of inputs; zero if no inputs
`count`	`int`	`0`	Number of inputs

2.32 Class: Multiplum

2.32.1 Inputs

Parameter	Type	Default	Description
`factors`	`vec_double`	`null`	Input factors

2.32.2 Outputs

Parameter	Type	Default	Description
`value`	`double`	`0.0`	The multiplum

2.33 Class: NumberedFile

2.33.1 Inputs

Parameter	Type	Default	Description
`fileName`	`string`		File name with extension, with/without a path
`evaluationOrder`	`int`	`0`	Number to append to file name
`numberWidth`	`int`	`0`	Number will be left-filled with zero up to this width

2.33.2 Outputs

Parameter	Type	Default	Description
`value`	`string`		File name with number

2.34 Class: OffsetDateTime

2.34.1 Inputs

Parameter	Type	Default	Description
`dateTime`	`datetime`	`null`	Date-time that will be offset
`days`	`int`	`0`	Offset (negative or positive)
`hours`	`int`	`0`	Offset (negative or positive)
`minutes`	`int`	`0`	Offset (negative or positive)

2.34.2 Outputs

Parameter	Type	Default	Description
`value`	`datetime`	`null`	Offset date-time

2.35 Class: OnOff

2.35.1 Inputs

Parameter	Type	Default	Description
`x`	`double`	`0.0`	x-value at which to calculate value
`xOn`	`double`	`0.0`	Lower threshold
`xOff`	`double`	`0.0`	Upper threshold
`valueOn`	`double`	`1.0`	Value inside [xOn,xOff] interval
`valueOff`	`double`	`0.0`	Value outside [xOn,xOff] interval
`isSticky`	`bool`	`FALSE`	Should remain on when switched on?

2.35.2 Outputs

Parameter	Type	Default	Description
`value`	`double`	`0.0`	Current on or off value
`isOn`	`bool`	`FALSE`	Is switch on?

2.36 Class: OnOffByDate

2.36.1 Inputs

Parameter	Type	Default	Description
x	date	null	x-value at which to calculate value
xOn	baredate	null	Lower threshold
xOff	baredate	null	Upper threshold
valueOn	double	1.0	Value inside [xOn,xOff] interval
valueOff	double	0.0	Value outside [xOn,xOff] interval
isSticky	bool	FALSE	Should remain on when switched on?

2.36.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Current on or off value
isOn	bool	FALSE	Is switch on?

2.37 Class: OutputR

2.37.1 Inputs

Parameter	Type	Default	Description
keepPlots	bool	TRUE	Keep previous plots before showing new plots in R?
clearMemory	bool	FALSE	Clear R memory?
showPlots	bool	TRUE	Show plots in R?
showLines	int	0	Number of lines to show at the prompt
popUp	bool	FALSE	Show plots in pop-up windows?
width	double	7.0	Width of pop-up windows (only used if popUp is set)
height	double	7.0	Height of pop-up windows (only used if popUp is set)
maximizeWindow	bool	FALSE	Maximize plot window size?
fontSize	double	0.0	Only used if >0
lineWidth	double	1.0	Width of lines in plots
plotAsList	bool	FALSE	Put plots into an R list?
saveDataFrame	bool	FALSE	Save output as R data frame?
skipFormats	bool	a)	
code	string		R code to be run before 'scripts'
scripts	vec_string	null	R scripts to be run at the end
plotTypes	vec_string	b)	
clearPlots	bool	FALSE	Deprecated
hasError	bool	/. [hasError]	

a) `OutputWriter::*[skipFormats]`

b) `PlotR::*[type]`

2.37.2 Outputs

Parameter	Type	Default	Description
ports	path	a)	Path to all ports used by my pages and plots
numPages	int	0	Number of pages in this output

a) `./PageR::*[xAxis]|./*/PlotR::*[ports]`

2.38 Class: OutputSelector

2.38.1 Inputs

Parameter	Type	Default	Description
beginStep	int	0	Output is written when this step is reached
beginDateTime	datetime	null	Output is written when this time point is reached (optional)
step	int	/.[step]	
dateTime	datetime	a)	
period	int	1	If >1: A row of output will be produced with this period
summary	string	"average"	Either "average" or "current"; how values are summarized over the given period
final	bool	FALSE	Overrules period
useLocalDecimalChar	bool	FALSE	Use local decimal character in output?
skipFormats	bool	b)	

a) `if exists(Calendar::*[dateTime]) then Calendar::*[dateTime] else .[beginDateTime]`
b) `! exists(OutputR::*[skipFormats])`

2.38.2 Outputs

Parameter	Type	Default	Description
isActive	bool	FALSE	Should output be written?
isSkipping	bool	FALSE	Are lines being skipped?

2.39 Class: OutputText

2.39.1 Inputs

Parameter	Type	Default	Description
ports	path	null	Path to port(s) to output

2.40 Class: OutputWriter

2.40.1 Inputs

Parameter	Type	Default	Description
showPorts	bool	FALSE	Shows a table of the ports to be written
ports	path	a)	
skipFormats	bool	b)	
useLocalDecimalChar	bool	c)	
isSkipping	bool	d)	

Parameter	Type	Default	Description
isActive	bool	e)	
period	int	f)	
summary	string	g)	

a) /.[iteration]|/.[step]|OutputText::*[ports]|OutputR::*[ports]
b) OutputSelector::*[skipFormats]
c) OutputSelector::*[useLocalDecimalChar]
d) OutputSelector::*[isSkipping]
e) OutputSelector::*[isActive]
f) OutputSelector::*[period]
g) OutputSelector::*[summary]

2.40.2 Outputs

Parameter	Type	Default	Description
filePath	string		Name of output file including absolute path
decimalChar	string		Decimal character used in output

2.41 Class: PageR

2.41.1 Inputs

Parameter	Type	Default	Description
xAxis	path	/.[step]	Port(s) on x-axis
title	string		Title shown on page
ncol	int	-1	No. of columns to arrange plots in
nrow	int	-1	No. of rows to arrange plots in
commonLegend	bool	FALSE	Collate legends of blots into one common legend
legendPosition	string	"bottom"	If 'commonLegend' then place legend here
width	double	..[width]	
height	double	..[height]	
maximizeWindow	bool	a)	
fontSize	double	..[fontSize]	
plotAsList	bool	b)	
popUp	bool	..[popUp]	
numPages	int	..[numPages]	
layouts	vec_string	c)	

a) ..[maximizeWindow]
b) ..[plotAsList]
c) ./PlotR::*[layout]

2.42 Class: Parasite

2.42.1 Inputs

Parameter	Type	Default	Description
density	double	0.0	Density
demand	double	0.0	Amount of resource needed

2.42.2 Outputs

Parameter	Type	Default	Description
supply	double	0.0	Total amount of resource acquired
sdRatio	double	0.0	Total supply/demand ratio

2.43 Class: PlotR

2.43.1 Inputs

Parameter	Type	Default	Description
ports	path	null	Port(s) on y-axis
layout	string	"facetted"	Either "merged" or "facetted"
type	string	"default"	Type of plot (default|density|histogram(nbins)|SobolConvergence|SobolIndices)
guideTitle	string		Title of guide legends
ggplot	string		R code that will be added to the ggplot
end	string		Deprecated
endCode	string		Deprecated
maxData	int	0	Max. number of data rows to plot; ignored if zero
ncol	int	-1	Number of columns in arrangement of plots; -1 keeps default
nrow	int	-1	Number of rows in arrangement of plots; -1 keeps default
direction	string	"row"	Fill in plots by 'row' or 'col'
iteration	int	a)	
xAxis	path	..[xAxis]	
width	double	..[width]	
height	double	..[height]	
fontSize	int	..[fontSize]	
plotAsList	bool	b)	

a) /.[iteration]
b) ..[plotAsList]

2.44 Class: PopUp

2.44.1 Inputs

Parameter	Type	Default	Description
title	string		Title

Parameter	Type	Default	Description
text	string		Short informative text
details	string		Further details
buttons	vec_string	c("OK")	Vector of buttons to show: (OK Yes No)
defaultButton	string	"OK"	Button assumed when Return is typed
escapeButton	string	"OK"	Button assumed when Esc is typed
icon	string	a)	Icon to show
when	string	"initialize"	In which step to show message, e.g., "initialize" or "debrief"
show	bool	TRUE	Determines whether pop-up will be shown

a) "information"

2.44.2 Outputs

Parameter	Type	Default	Description
answer	string		Name of button pushed
accepted	bool	FALSE	Question accepted?

2.45 Class: Population

2.45.1 Inputs

Parameter	Type	Default	Description
initial	double	100.0	
ageIncrement	double	0.0	
bufferSize	int	100	
firstCohortGain	double	0.0	
cohortsGain	vec_double	null	
cohortsLoss	vec_double	null	

2.45.2 Outputs

Parameter	Type	Default	Description
lastCohortSpill	double	0.0	
cohorts	vec_double	null	
age	vec_double	null	

2.46 Class: Predator

2.46.1 Inputs

Parameter	Type	Default	Description
density	double	0.0	Density
demand	double	0.0	Amount of resource needed

2.46.2 Outputs

Parameter	Type	Default	Description
supply	double	0.0	Total amount of resource acquired

Parameter	Type	Default	Description
sdRatio	double	0.0	Total supply/demand ratio

2.47 Class: Prey

2.47.1 Inputs

Parameter	Type	Default	Description
density	double	0.0	Density
attackRate	double	0.0	Attack rate
gainFactor	double	1.0	Conversion of resource into supply units

2.47.2 Outputs

Parameter	Type	Default	Description
loss	double	0.0	Amount of prey lost to predators
mortality	double	0.0	Mortality caused by predators

2.48 Class: PrioritySignal

2.48.1 Inputs

Parameter	Type	Default	Description
initialSignal	double	0.0	Value of signal at reset
initialFlag	bool	FALSE	Value of flag at reset
myFlags	vec_bool	a)	
mySignals	vec_double	b)	
reverseOrder	bool	FALSE	Find first signal!=0 from top (false) or bottom (true)?

a) ./BaseSignal::*[flagIsUp]
b) ./BaseSignal::*[signal]

2.48.2 Outputs

Parameter	Type	Default	Description
signal	double	0.0	Value of the signal
flagIsUp	bool	FALSE	Is the signal on?
flagJustRaised	bool	FALSE	Was the signal just set on?
flagJustLowered	bool	FALSE	Was the signal just set off?
value	double	0.0	Synonym for signal

2.49 Class: ProportionalSignal

2.49.1 Inputs

Parameter	Type	Default	Description
initialSignal	double	0.0	Value of signal at reset
initialFlag	bool	FALSE	Value of flag at reset
input	double	25.0	Value determining the signal

Parameter	Type	Default	Description
threshold	double	25.0	Input threshold above which the signal is changing
thresholdBand	double	5.0	Interval of the input over which the signal is changing
minSignal	double	0.0	Minimum possible signal
maxSignal	double	100.0	Maximum possible signal
increasingSignal	bool	TRUE	Is the signal increasing inside the threshold band?

2.49.2 Outputs

Parameter	Type	Default	Description
signal	double	0.0	Value of the signal
flagIsUp	bool	FALSE	Is the signal on?
flagJustRaised	bool	FALSE	Was the signal just set on?
flagJustLowered	bool	FALSE	Was the signal just set off?
value	double	0.0	Synonym for signal

2.50 Class: RandomBinomial

2.50.1 Inputs

Parameter	Type	Default	Description
P	double	0.5	Probability of event
drawAtInitialize	bool	a)	
drawAtReset	bool	b)	
drawAtUpdate	bool	c)	
min	bool	FALSE	Not used
max	bool	TRUE	Not used

a) ancestors::*/RandomiserBase::*[drawAtInitialize]
b) ancestors::*/RandomiserBase::*[drawAtReset]
c) ancestors::*/RandomiserBase::*[drawAtUpdate]

2.50.2 Outputs

Parameter	Type	Default	Description
value	bool	FALSE	The most recently drawn value

2.51 Class: RandomLogNormal

2.51.1 Inputs

Parameter	Type	Default	Description
P	double	0.95	Range of the distribution covered by the [minValue;maxValue] range
drawAtInitialize	bool	a)	
drawAtReset	bool	b)	
drawAtUpdate	bool	c)	

Parameter	Type	Default	Description
min	double	0.01	Minimum value (included)
max	double	1.0	Maximum value (excluded)

a) ancestors::*/RandomiserBase::*[drawAtInitialize]
b) ancestors::*/RandomiserBase::*[drawAtReset]
c) ancestors::*/RandomiserBase::*[drawAtUpdate]

2.51.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	The most recently drawn value

2.52 Class: RandomLogUniform

2.52.1 Inputs

Parameter	Type	Default	Description
P	double	0.95	Range of the distribution covered by the [minValue;maxValue] range
drawAtInitialize	bool	a)	
drawAtReset	bool	b)	
drawAtUpdate	bool	c)	
min	double	0.01	Minimum value (included)
max	double	1.0	Maximum value (excluded)

a) ancestors::*/RandomiserBase::*[drawAtInitialize]
b) ancestors::*/RandomiserBase::*[drawAtReset]
c) ancestors::*/RandomiserBase::*[drawAtUpdate]

2.52.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	The most recently drawn value

2.53 Class: RandomNormal

2.53.1 Inputs

Parameter	Type	Default	Description
P	double	0.95	Range of the distribution covered by the [minValue;maxValue] range
drawAtInitialize	bool	a)	
drawAtReset	bool	b)	
drawAtUpdate	bool	c)	
min	double	0.0	Minimum value (included)
max	double	1.0	Maximum value (excluded)

a) ancestors::*/RandomiserBase::*[drawAtInitialize]
b) ancestors::*/RandomiserBase::*[drawAtReset]
c) ancestors::*/RandomiserBase::*[drawAtUpdate]

2.53.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	The most recently drawn value

2.54 Class: RandomUniform

2.54.1 Inputs

Parameter	Type	Default	Description
P	double	0.95	Range of the distribution covered by the [minValue;maxValue] range
drawAtInitialize	bool	a)	
drawAtReset	bool	b)	
drawAtUpdate	bool	c)	
min	double	0.0	Minimum value (included)
max	double	1.0	Maximum value (excluded)

a) ancestors::*/RandomiserBase::*[drawAtInitialize]
b) ancestors::*/RandomiserBase::*[drawAtReset]
c) ancestors::*/RandomiserBase::*[drawAtUpdate]

2.54.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	The most recently drawn value

2.55 Class: RandomUniformInt

2.55.1 Inputs

Parameter	Type	Default	Description
P	double	0.95	Range of the distribution covered by the [minValue;maxValue] range
drawAtInitialize	bool	a)	
drawAtReset	bool	b)	
drawAtUpdate	bool	c)	
min	int	0	Minimum value (included)
max	int	1	Maximum value (included)

a) ancestors::*/RandomiserBase::*[drawAtInitialize]
b) ancestors::*/RandomiserBase::*[drawAtReset]
c) ancestors::*/RandomiserBase::*[drawAtUpdate]

2.55.2 Outputs

Parameter	Type	Default	Description
value	int	0	The most recently drawn value

2.56 Class: RandomiserMonteCarlo

2.56.1 Inputs

Parameter	Type	Default	Description
iteration	int	a)	
iterations	int	b)	
doSensitivityAnalysis	bool	FALSE	Carry out a sensitivity analysis?
bootstrapSize	int	1000	Size of bootstrap sample (cheap in computation time); only used in sensitivity analysis
seed	int	0	Seed for random numbers; if this is zero a random seed value will be used
drawAtInitialize	bool	FALSE	Draw a value when a box is initialized?
drawAtReset	bool	TRUE	Draw a value when a box is reset?
drawAtUpdate	bool	FALSE	Draw a value when a box is updated?

a) /.[iteration]
b) /.[iterations]

2.56.2 Outputs

Parameter	Type	Default	Description
numVariables	int	0	Number of variables which are randomised

2.57 Class: RandomiserSobolSequence

2.57.1 Inputs

Parameter	Type	Default	Description
iteration	int	a)	
iterations	int	b)	
doSensitivityAnalysis	bool	FALSE	Carry out a sensitivity analysis?
bootstrapSize	int	1000	Size of bootstrap sample (cheap in computation time); only used in sensitivity analysis
seed	int	0	Seed for random numbers; if this is zero a random seed value will be used
drawAtInitialize	bool	FALSE	Draw a value when a box is initialized?
drawAtReset	bool	TRUE	Draw a value when a box is reset?
drawAtUpdate	bool	FALSE	Draw a value when a box is updated?

a) /.[iteration]
b) /.[iterations]

2.57.2 Outputs

Parameter	Type	Default	Description
numVariables	int	0	Number of variables which are randomised

2.58 Class: RandomiserStratified

2.58.1 Inputs

Parameter	Type	Default	Description
iteration	int	a)	
iterations	int	b)	
doSensitivityAnalysis	bool	FALSE	Carry out a sensitivity analysis?
bootstrapSize	int	1000	Size of bootstrap sample (cheap in computation time); only used in sensitivity analysis
seed	int	0	Seed for random numbers; if this is zero a random seed value will be used
drawAtInitialize	bool	FALSE	Draw a value when a box is initialized?
drawAtReset	bool	TRUE	Draw a value when a box is reset?
drawAtUpdate	bool	FALSE	Draw a value when a box is updated?

a) `/.[iteration]`
b) `/.[iterations]`

2.58.2 Outputs

Parameter	Type	Default	Description
numVariables	int	0	Number of variables which are randomised

2.59 Class: Ratio

2.59.1 Inputs

Parameter	Type	Default	Description
denominator	double	0.0	Number to be divided
divisor	double	0.0	The divisor
zeroTolerance	double	0.0	Used for infinity check
allowInfinityResult	bool	TRUE	Will yield an error if `divisor < zeroTolerance`

2.59.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	The ratio

2.60 Class: Records

2.60.1 Inputs

Parameter	Type	Default	Description
fileName	string	a)	Name of file with records; columns separated by white space
dateColumnName	string	"Date"	Name of column with date
timeColumnName	string	"Time"	Name of column with time
cycle	bool	FALSE	Cycle back to start at end of file? Forces ignoreYear=true

Parameter	Type	Default	Description
ignoreYear	bool	FALSE	Ignore year when synchronising with the calendar
calendarDateTime	datetime	b)	

a) "records.txt"
b) calendar[dateTime]

2.60.2 Outputs

Parameter	Type	Default	Description
firstDateTime	datetime	null	Date-time stamp of the first line in the file
lastDateTime	datetime	null	Date-time stamp of the last line in the file
currentDateTime	datetime	null	Date-time stamp of the current outputs
nextDateTime	datetime	null	Date-time stamp of the next outputs
currentDate	date	null	Date stamp of the current outputs
nextDate	date	null	Date stamp of the next outputs
currentTime	time	null	Time stamp of the current outputs
nextTime	time	null	Time stamp of the next outputs

2.61 Class: RunningAverageSteps

2.61.1 Inputs

Parameter	Type	Default	Description
initial	double	0.0	Value at reset
input	double	0.0	The input to sum or average
count	int	1	Number of steps to average over

2.61.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Running sum or average

2.62 Class: RunningAverageTimed

2.62.1 Inputs

Parameter	Type	Default	Description
initial	double	0.0	Value at reset
input	double	0.0	The input to sum or average
timeWindow	double	1.0	Time window to average over in units of timeUnit
timeUnit	string	"d"	Unit of time window
timeStepSecs	double	a)	

a) calendar[timeStepSecs]

2.62.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Running sum or average
count	int	0	Number of steps averages over

2.63 Class: RunningSumSteps

2.63.1 Inputs

Parameter	Type	Default	Description
initial	double	0.0	Value at reset
input	double	0.0	The input to sum or average
count	int	1	Number of steps to average over

2.63.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Running sum or average

2.64 Class: RunningSumTimed

2.64.1 Inputs

Parameter	Type	Default	Description
initial	double	0.0	Value at reset
input	double	0.0	The input to sum or average
timeWindow	double	1.0	Time window to average over in units of timeUnit
timeUnit	string	"d"	Unit of time window
timeStepSecs	double	a)	

a) `calendar[timeStepSecs]`

2.64.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Running sum or average
count	int	0	Number of steps averages over

2.65 Class: Sah

2.65.1 Inputs

Parameter	Type	Default	Description
temperature	double	0.0	Temperature
value	double	0.0	Saturated absolute humidity

2.66 Class: Sandy

2.66.1 Inputs

Parameter	Type	Default	Description
xMin	double	0.0	

Parameter	Type	Default	Description
xMax	double	1.0	
yMax	double	1.0	
a	double	1.0	
b	double	1.0	
x	double	0.0	

2.66.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	

2.67 Class: Scenarios

2.67.1 Inputs

Parameter	Type	Default	Description
fileName	string	a)	Text file with columns, one for each output produced. One iteration is run for each line
title	string	"Scenario"	Name of column that holds title of scenario (optional)

a) "scenarios.txt"

2.67.2 Outputs

Parameter	Type	Default	Description
iterations	int	0	Number of iterations needed to run scenarios

2.68 Class: SelectFile

2.68.1 Inputs

Parameter	Type	Default	Description
folder	string		Relative path to folder
filter	string	"*.txt"	File filter
selectFileNumber	int	1	Which file to pick
fileNumberOffset	int	1	Offset for fileNumber, typically 0 or 1
showFileNames	bool	TRUE	Show set of file names?

2.68.2 Outputs

Parameter	Type	Default	Description
numFiles	int	0	Number of matching files
fileName	string		Name of selected file
fileNamePath	string		Name of selected file with absolute path

2.69 Class: SequenceByIteration

2.69.1 Inputs

Parameter	Type	Default	Description
counter	int	0	Current counter into values
min	double	0.0	Minimum value in sequence
max	double	1.0	Maximum value in sequence
by	double	0.1	Maximum value in sequence

2.69.2 Outputs

Parameter	Type	Default	Description
numValues	int	0	Number of values
values	vec_double	null	Vector of all values
value	double	0.0	Current value
iterations	int	0	The number of iterations needed to run through all values

2.70 Class: SequenceByStep

2.70.1 Inputs

Parameter	Type	Default	Description
counter	int	0	Current counter into values
min	double	0.0	Minimum value in sequence
max	double	1.0	Maximum value in sequence
by	double	0.1	Maximum value in sequence

2.70.2 Outputs

Parameter	Type	Default	Description
numValues	int	0	Number of values
values	vec_double	null	Vector of all values
value	double	0.0	Current value
steps	int	0	The number of steps needed to run through all values

2.71 Class: SeriesByIteration

2.71.1 Inputs

Parameter	Type	Default	Description
counter	int	0	Current counter into values
numbers	vec_double	null	Vector holding a series of values

2.71.2 Outputs

Parameter	Type	Default	Description
numValues	int	0	Number of values
values	vec_double	null	Vector of all values

Parameter	Type	Default	Description
value	double	0.0	Current value
iterations	int	0	The number of iterations needed to run through all values

2.72 Class: SeriesByStep

2.72.1 Inputs

Parameter	Type	Default	Description
counter	int	0	Current counter into values
numbers	vec_double	null	Vector holding a series of values

2.72.2 Outputs

Parameter	Type	Default	Description
numValues	int	0	Number of values
values	vec_double	null	Vector of all values
value	double	0.0	Current value
steps	int	0	The number of steps needed to run through all values

2.73 Class: Simulation

2.73.1 Inputs

Parameter	Type	Default	Description
iterations	int	1	Number of iterations to run
steps	int	a)	
stopIteration	bool	FALSE	Stop running after this iteration?
stopStep	bool	FALSE	Stop running after this step?
silent	bool	FALSE	Run without scrolling progress messages?
unattended	bool	FALSE	An unattended run will finish without accessing clipboard

a) `if exists(Calendar::*[steps]) then Calendar::*[steps] else 1`

2.73.2 Outputs

Parameter	Type	Default	Description
iteration	int	0	Iteration number (1,2,...)
step	int	0	Step number (0,1,2,...)
finalStep	int	0	Last completed step in simulation
executionTime	int	0	Duration of simulation run (ms)
hasRun	bool	FALSE	Has this simulation been run?
hasError	bool	FALSE	Did an error occur during simulation run?
errorMsg	string		Error message

2.74 Class: SineWaveTemperature

2.74.1 Inputs

Parameter	Type	Default	Description
recordsName	string	"weather"	Name of Records box holding temperature
TminName	string	"Tmin"	Name of column holding daily min. temperature
TmaxName	string	"Tmax"	Name of column holding daily max. temperature
morningOffset	double	0.5	Hours after sunrise when temperature is at minimum
afternoonOffset	double	3.0	Hours after noon when temperature is at maximum
time	time	a)	
sunrise	time	b)	
solarTime	time	c)	
timeStep	int	d)	
timeUnit	string	e)	

a) calendar[time]
b) calendar[sunrise]
c) calendar[trueSolarTime]
d) calendar[timeStep]
e) calendar[timeUnit]

2.74.2 Outputs

Parameter	Type	Default	Description
TminYesterday	double	0.0	Yesterday's Tmin
TmaxYesterday	double	0.0	Yesterday's Tmax
TminToday	double	0.0	Today's Tmin
TmaxToday	double	0.0	Today's Tmax
TminTomorrow	double	0.0	Tomorrow's Tmin
TmaxTomorrow	double	0.0	Tomorrow's Tmax
value	double	0.0	Sine-interpolated temperature

2.75 Class: SobolGFunction

2.75.1 Inputs

Parameter	Type	Default	Description
a	vec_double	null	Vector of parameters
x	vec_double	null	Vector of inputs

2.75.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	B value

2.76 Class: Split

2.76.1 Inputs

Parameter	Type	Default	Description
whole	double	0.0	The number to be split
proportions	vec_double	null	Vector of proportions; sum<=1

2.76.2 Outputs

Parameter	Type	Default	Description
pieces	vec_double	null	Vector holding proportions of whole; same length as proportions

2.77 Class: Stage

2.77.1 Inputs

Parameter	Type	Default	Description
initial	double	0.0	Initial amount of inflow
k	int	30	Distribution parameter; use small k for larger dispersion in output
duration	double	100.0	Average delay between inflow and outflow
timeStep	double	1.0	Time step
growthFactor	double	1.0	Factor by which outflow will be scaled relative to inflow
instantLossRate	double	0.0	Mortality applied before inflow is added [0..1]
phaseInflow	vec_double	null	Phase inflow cohorts (vector)
zeroLimit	double	1e-12	Cut contents to zero when total is less than this
inflow	double	0.0	Amount of inflow
phaseOutflowProportion	double	0.0	Proportion that will change phase in next time step

2.77.2 Outputs

Parameter	Type	Default	Description
content	double	0.0	Total content inside the stage
value	double	0.0	Synonym for 'content'
inflowSum	double	0.0	Sum of current stage inflow (scalar)
inflowTotal	double	0.0	Accumulated total stage inflow (scalar)
phaseInflowSum	double	0.0	Sum of current phase inflow cohorts (scalar)
phaseInflowTotal	double	0.0	Accumulated total phase inflow (scalar)
outflowSum	double	0.0	Sum of current stage outflow (scalar)
outflowTotal	double	0.0	Accumulated total stage outflow (scalar)
phaseOutflowSum	double	0.0	Sum of current phase outflow cohorts (scalar)
phaseOutflowTotal	double	0.0	Accumulated total phase outflow (scalar)
phaseOutflow	vec_double	null	Phase outflow cohorts (vector)
growth	double	0.0	Change in content+outflow during this time step
outflow	double	0.0	Outflow emerging from the stage

2.78 Class: StageAndPhase

2.78.1 Inputs

Parameter	Type	Default	Description
initial	double	0.0	Initial amount of inflow
k	int	30	Distribution parameter; use small k for larger dispersion in output
duration	double	100.0	Average delay between inflow and outflow
timeStep	double	1.0	Time step
growthFactor	double	1.0	Factor by which outflow will be scaled relative to inflow
instantLossRate	double	0.0	Mortality applied before inflow is added [0..1]
phaseInflow	vec_double	null	Phase inflow cohorts (vector)
zeroLimit	double	1e-12	Cut contents to zero when total is less than this
inflow	vec_double	null	Amount of inflow (vector)
scalarInflow	double	0.0	Amount of inflow (scalar)
phaseK	int	30	Phase distribution parameter
phaseDuration	double	100.0	Average delay between inflow and outflow through phase
phaseTimeStep	double	1.0	Time step for phase development

2.78.2 Outputs

Parameter	Type	Default	Description
content	double	0.0	Total content inside the stage
value	double	0.0	Synonym for 'content'
inflowSum	double	0.0	Sum of current stage inflow (scalar)
inflowTotal	double	0.0	Accumulated total stage inflow (scalar)
phaseInflowSum	double	0.0	Sum of current phase inflow cohorts (scalar)
phaseInflowTotal	double	0.0	Accumulated total phase inflow (scalar)
outflowSum	double	0.0	Sum of current stage outflow (scalar)
outflowTotal	double	0.0	Accumulated total stage outflow (scalar)
phaseOutflowSum	double	0.0	Sum of current phase outflow cohorts (scalar)
phaseOutflowTotal	double	0.0	Accumulated total phase outflow (scalar)
phaseOutflow	vec_double	null	Phase outflow cohorts (vector)
growth	double	0.0	Change in content+outflow during this time step
outflow	vec_double	null	Stage outflow cohorts (vector)

2.79 Class: Sum

2.79.1 Inputs

Parameter	Type	Default	Description
values	vec_double	null	Input values

2.79.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Sum of inputs; zero if no inputs
count	int	0	Number of inputs

2.80 Class: Sun

2.80.1 Inputs

Parameter	Type	Default	Description
latitude	double	a)	
longitude	double	b)	
dateTime	datetime	c)	
timeZone	double	d)	
altitude	double	0.0	Altitude above sea level

a) calendar[latitude]
b) calendar[longitude]
c) calendar[dateTime]
d) calendar[timeZone]

2.80.2 Outputs

Parameter	Type	Default	Description
sunrise	time	null	Time of sunrise
noon	time	null	Time of nonw
sunset	time	null	Time of sunset
solarTime	time	null	Solar time (noon=12:00:00
dayLength	double	0.0	Day length from sunrise to sunset
elevation	double	0.0	Solar angle to horizon
azimuth	double	0.0	Compass direction to sun (north=0)
isDay	bool	FALSE	Is the sun up?
isNight	bool	FALSE	Has the sun set?
solarConstant	double	0.0	The irradiation at the top of the atmosphere
angot	double	0.0	The irradiation on Earth surface under optimal atmospheric conditions

2.81 Class: SunDiffuseRadiation

2.81.1 Inputs

Parameter	Type	Default	Description
globalRadiation	double	0.0	Global radiation
angotRadiation	double	sun[angot]	

2.81.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Proportion diffuse light

2.82 Class: SuperFunctionalResponse

2.82.1 Inputs

Parameter	Type	Default	Description
attacker	double	0.0	Attacker density
host	double	0.0	Resource density
demand	double	0.0	Number of attacks demanded
attackRate	double	0.0	#attacks per #host per #attacker per time unit
timeStep	double	1.0	Time step

2.82.2 Outputs

Parameter	Type	Default	Description
supply	double	0.0	Number of attacks accomplished
hostsAttacked	double	0.0	The number of hosts attacked
propHostsAttacked	double	0.0	Proportion of hosts attacked
propAttackersAttacked	double	0.0	Proportion of successful attackers
sdRatio	double	0.0	Attackers's supply/demand ratio

2.83 Class: SupplyBudget

2.83.1 Inputs

Parameter	Type	Default	Description
supplyGross	double	0.0	Gross supply
demandNet	double	0.0	Net demand
demandResp	double	0.0	Respiration demand
egested	double	0.0	Egested or unused fraction of prey
conversionCost	double	0.0	Cost of converting prey into consumer)

2.83.2 Outputs

Parameter	Type	Default	Description
supplyNet	double	0.0	Net supply for predator assimilation
egestedAmount	double	0.0	Egested (or unused) amount of resource
conversionCostAmount	double	0.0	Cost of converting resource into consumer
sdRatioNet	double	0.0	Predator's net supply/demand ratio

2.84 Class: ThresholdSignal

2.84.1 Inputs

Parameter	Type	Default	Description
initialSignal	double	0.0	Value of signal at reset
initialFlag	bool	FALSE	Value of flag at reset
input	double	0.0	Input that is compared to threshold
threshold	double	0.0	Threshold value
flagAbove	bool	TRUE	Raise flag when input is at or above threshold?

Parameter	Type	Default	Description
signalFlagged	double	1.0	Signal when flagged
signalUnflagged	double	0.0	Signal when not flagged

2.84.2 Outputs

Parameter	Type	Default	Description
signal	double	0.0	Value of the signal
flagIsUp	bool	FALSE	Is the signal on?
flagJustRaised	bool	FALSE	Was the signal just set on?
flagJustLowered	bool	FALSE	Was the signal just set off?
value	double	0.0	Synonym for signal

2.85 Class: TimeToHours

2.85.1 Inputs

Parameter	Type	Default	Description
time	time	00:00:00	Time of day

2.85.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Hours since midnight

2.86 Class: VectorScaled

2.86.1 Inputs

Parameter	Type	Default	Description
vector	vec_double	null	Vector to be scaled
factor	double	0.0	Scaling factor

2.86.2 Outputs

Parameter	Type	Default	Description
value	vec_double	null	Scaled vector

2.87 Class: VectorSum

2.87.1 Inputs

Parameter	Type	Default	Description
input	string		Path to vectors (must have same length)

2.87.2 Outputs

Parameter	Type	Default	Description
value	vec_double	null	Sum; a vector of same length as the inputs

2.88 Class: boxesDocumentation

2.88.1 Inputs

Parameter	Type	Default	Description
title	string	a)	
authorNames	vec_string	null	
authorInstitutes	vec_string	null	
authorEmails	vec_string	null	

a) Basic building blocks

3 Module: callosobruchus

3.1 Class: DensityDependent

3.1.1 Inputs

Parameter	Type	Default	Description
seedCapacity	int	9	Number of beetles one seed can sustain
beetlesProducedTotal	double	0.0	Total number of beetles produced since stocking
seedsTotal	double	0.0	Total number of seeds stocked
k	double	8.17	Dispersion of the negative binomial
exponent	double	-0.496	Negative exponential response

3.1.2 Outputs

Parameter	Type	Default	Description
avgBeetlesProducedPerSeed	double	0.0	Average number of eggs per seed, laid since stocking
survival	double	0.0	Survival through juvenile life stages
damage	double	0.0	Proportion of seeds damaged

3.2 Class: Fecundity

3.2.1 Inputs

Parameter	Type	Default	Description
sexRatio	double	0.5	Proportion of females
temperature	double	0.0	Store temperature
slope	double	1.279	Fecundity response on temperature
intercept	double	-23.1	Fecundity response on temperature
minTemperature	double	25.0	No extrapolation below this temperature
maxTemperature	double	40.0	No extrapolation above this temperature

3.2.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Total eggs laid per female

3.3 Class: Stock

3.3.1 Inputs

Parameter	Type	Default	Description
size	double	100.0	Weight of seeds stored
seedsPerKg	double	7000.0	Number of seeds per kg put on storage

3.3.2 Outputs

Parameter	Type	Default	Description
seedsTotal	double	0.0	Total no. of seeds

3.4 Class: callosobruchusDocumentation

3.4.1 Inputs

Parameter	Type	Default	Description
title	string	a)	
authorNames	vec_string	null	
authorInstitutes	vec_string	null	
authorEmails	vec_string	null	

a) callosobruchus model

4 Module: cmbp

4.1 Class: DevRate

4.1.1 Inputs

Parameter	Type	Default	Description
b1	double	0.0	Regression parameter
b2	double	0.0	Regression parameter
b3	double	0.0	Regression parameter
b4	double	0.0	Regression parameter
T	double	0.0	Ambient temperature (oC)

4.1.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Daily development rate

4.2 Class: Emigration

4.2.1 Inputs

Parameter	Type	Default	Description
e2	double	0.0	Regression parameter
e3	double	0.0	Regression parameter
N	double	0.0	Adult density

4.2.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Peak curve

4.3 Class: ExponentialPeak

4.3.1 Inputs

Parameter	Type	Default	Description
d1	double	0.0	Regression parameter
d2	double	0.0	Regression parameter
x	double	0.0	Independent variable
optimum	double	0.0	x-value at peak y

4.3.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Peak curve

4.4 Class: Fecundity

4.4.1 Inputs

Parameter	Type	Default	Description
maxFecundity	double	700.0	Life-time fecundity of a female at 30oC
r	double	0.2359	Exponential increase in fecundity with temperature
T	double	0.0	Store temperature (oC)
sexRatio	double	0.5	Sex ratio [0;1]

4.4.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Life-time fecundity scaled by temperature

4.5 Class: GrainMoistureContent

4.5.1 Inputs

Parameter	Type	Default	Description
c1	double	0.523	Regression parameter
c2	double	-0.00967	Regression parameter
c3	double	7.06e-05	Regression parameter
rh	double	0.0	Ambient relative humidity (%)

4.5.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Grain moisture content (%)

4.6 Class: GrainStore

4.6.1 Inputs

Parameter	Type	Default	Description
Tmin	double	0.0	Outdoors daily minimum temperature (oC)
Tmax	double	0.0	Outdoors daily maximum temperature (oC)
Toffset	double	0.0	Offset for temperature inside the store (oC)
mass	double	500.0	Mass of stored grain (kg)

4.6.2 Outputs

Parameter	Type	Default	Description
T	double	0.0	Store temperature (oC

4.7 Class: Immigration

4.7.1 Inputs

Parameter	Type	Default	Description
number	double	0.0	Nuber of P.t. arriving
storeMass	double	500.0	Size of store (kg maize)
day	int	0	Day of immigration (no. of days after start)

4.7.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Immigrant increase in P.t. density (per kg

4.8 Class: SurvivalByDensity

4.8.1 Inputs

Parameter	Type	Default	Description
e1	double	0.0	Regression parameter
N	double	0.0	Adult density

4.8.2 Outputs

Parameter	Type	Default	Description
value	double	0.0	Peak curve

5 Module: command

5.1 Class: commandDocumentation

5.1.1 Inputs

Parameter	Type	Default	Description
title	string	a)	
authorNames	vec_string	null	
authorInstitutes	vec_string	null	
authorEmails	vec_string	null	

a) Commands for the prompt